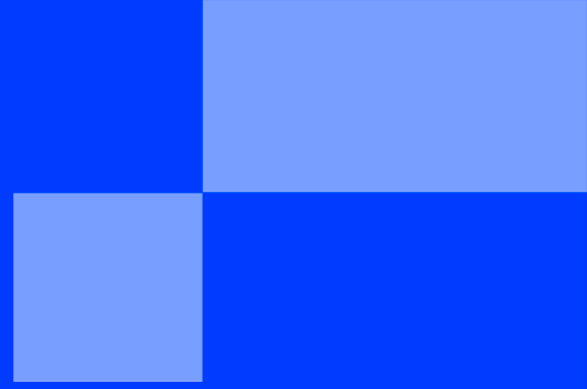




SUMMER SCHOOL

< PART 1 BOOTSTRAP - THE ART OF
PROMPTING />

SUMMER SCHOOL

< AI & AUTOMATION: UPGRADING YOUR DAILY LIFE />

"Any sufficiently advanced technology is indistinguishable from magic."
— **Arthur C. Clarke**

Introduction

Welcome to the Artificial Intelligence module! Forget everything you know about coding for a moment. You do not need to be a math genius or a programmer to use AI.

Right now, most people use tools like ChatGPT or Gemini like a traditional search engine. This is a mistake.

An AI is not a search engine, it is a digital intern.

If you give a human intern a vague instruction like "Write an email to my boss," you will get a bad result. If you give an AI a vague instruction, you will also get a bad result.

In this bootstrap, you will learn the secret to unlocking the true power of LLMs: **Prompt Engineering**. We will use real-life examples to upgrade your daily productivity.

Step 1: The Anatomy of a Perfect Prompt

A "Prompt" is the text you type to talk to the AI. To get professional results, you must stop typing single sentences. A perfect prompt always contains 4 elements (the **CTRF** Framework):

1. **Context:** What is the background situation?
2. **Task:** What exactly do you want the AI to do?
3. **Role (Persona):** Who should the AI pretend to be?
4. **Format:** How should the answer look?

Step 2: Lab 1 - The Ultimate Study Buddy

Imagine you just came out of a chaotic 2-hour lecture. Your notes are a mess. Let's use AI to save your evening.

Open [ChatGPT](#), [Claude](#), or [Gemini](#) and try this weak prompt:

> Summarize these notes for me: *[Insert some random messy text here]*

The result will be okay, but boring. Now, let's use the **CTRF framework**. Open a new chat and copy-paste this exact prompt:

[Role] You are an expert university tutor who is amazing at helping beginners understand complex topics.

[Context] I am a first-year student. I just took some very messy notes during my History of Computing class, and my exam is tomorrow. I am stressed.

[Task] Please clean up my notes, fix the spelling mistakes, and explain any complicated words simply.

[Format] Output a beautiful study guide. Use bold headers, bullet points, and add a "TL;DR" (Too Long; Didn't Read) summary of 3 sentences at the very top. Add relevant emojis.

Here are my notes:

"computers started huge like the ENIAC room size then transistors made them smaller in the 50s i think? then microprocessors intel 4004 allowed personal computers to exist apple and microsoft started in the 70s internet came from arpanet military stuff now we have smart-phones which are pocket computers, don't forget about AI and machine learning which is like teaching computers to learn from data, neural networks are inspired by the brain, quantum computing is the future maybe."

Action: Look at the difference! By giving the AI a role and a format, it transformed from a basic robot into a dedicated teacher.

Step 3: Lab 2 - The Diplomat (Tone Control)

AI is incredibly good at adjusting its "tone of voice". This is a superpower for writing difficult emails. Imagine your rent is due tomorrow, but you haven't received your paycheck yet. You need to ask your landlord for a 5-day delay.

Try this prompt:

- > **[Role]** You are an expert in professional communication and conflict resolution.
- > **[Context]** I need to email my landlord, Mr. Smith. I have lived in his apartment for 2 years and always paid on time. However, due to a bank error, my salary will be 5 days late this month.
- > **[Task]** Write an email to Mr. Smith asking if I can pay my rent on the 5th instead of the 1st.
- > **[Format]** The email must be short, highly respectful, reassuring, and professional.

Action: Read the email. Notice how polite it is.

Now, let's change the **Tone**. Reply to the AI with this:

- > *Actually, Mr. Smith is a terrible landlord. He hasn't fixed my heater in 3 months despite my calls. Rewrite the email. The tone must now be firm, slightly passive-aggressive, and mention that I am holding the rent as leverage until the heater is fixed.*

Action: Watch how the AI completely pivots its personality and vocabulary based purely on your tone instructions!

Conclusion

You now possess the core skills of Prompt Engineering:

- You know how to assign a **Role**.
- You know how to provide **Context**.
- You know how to enforce a specific **Format**.

Close your playground. It's time to open the main Project document. Your Web Development team needs content, and you are going to use these exact techniques to become an automated journalist!



Pro Tip: The more specific your instructions, the better the AI will perform. If you want to get a perfect result, you must be very clear about what you want. The AI is not a mind reader. It will only do what you tell it to do.



Hallucination: The AI might sometimes generate information that seems plausible but is actually false or misleading. Always verify the AI's output for accuracy. You can use the AI itself to fact-check its own output by asking it to provide sources or references for the information it generates.

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